



**SYDNEY BOYS HIGH
SCHOOL**
MOORE PARK, SURRY HILLS

2005

YEAR 12

ASSESSMENT TASK #2

Mathematics Extension 1

General Instructions

- Working time – 90 minutes.
- Reading Time – 5 minutes.
- Write using black or blue pen.
- Board approved calculators may be used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may not be awarded for messy or badly arranged work
- Hand in your answer booklets in 3 sections. Section A (Question 1), Section B (Question 2) and Section C (Question 3)

Total Marks - 76

- Attempt questions 1 – 3
- All sections are **NOT** of equal value.

Examiner: *A. Fuller*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

Total marks - 76

Attempt Questions 1 - 3

All questions are NOT of equal value

Answer each SECTION in a SEPARATE writing booklet.

Section A

Marks

Question 1 (22 marks)

(a) Differentiate the following:

i) $\tan 2x$ 1

ii) $\frac{1}{e^{\frac{x}{2}}}$ 2

iii) $\ln(1-2x)^2$ 2

iv) $\sin^2(3x)$ 2

(b) Find the following:

i) $\int \cos \frac{x}{2} dx$ 1

ii) $\int e^{1-4x} dx$ 1

iii) $\int \cot x dx$ 2

iv) $\int 3x.e^{x^2} dx$ 2

(c) Find $\lim_{x \rightarrow 0} \frac{\sin 2x}{x}$ 1

(d) i) Show that the equation $x \cdot \ln x - 1 = 0$ has a root between $x = 1$ and $x = 2$. 2

ii) Using $x = 2$ as the first approximation, apply Newton's method once to obtain a better approximation of this root. (correct to two decimal places) 2

(e) i) Show that $\frac{2x+1}{x+2} = 2 - \frac{3}{x+2}$ 1

ii) Hence or otherwise, find the exact value of $\int_0^1 \frac{2x+1}{x+2} dx$. 3

End of Section A

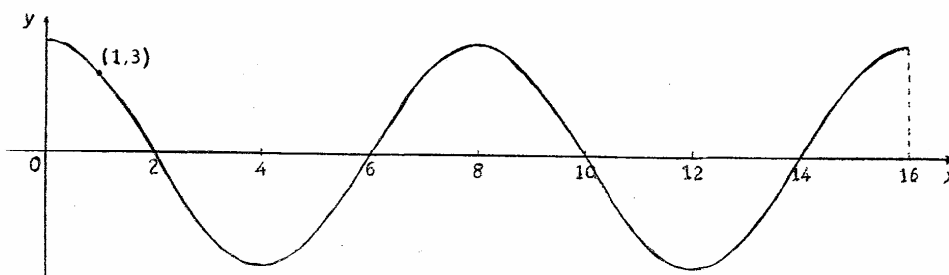
Section B (Use a SEPARATE writing booklet)

Marks

Question 2 (28 marks)

- (a) Consider the function $f(x) = \log_e(3x-6)$
- | | | |
|------|---|----------|
| i) | State the largest possible domain of $f(x)$. | 1 |
| ii) | Sketch the curve $y = f(x)$. | 2 |
| iii) | Find the equation of the normal to the curve at the point where $x = 4$ | 3 |

(b)

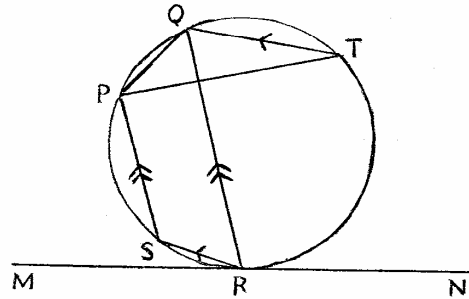


Given that the equation of the above curve is of the form $y = a \cos(bx)$ find the values of a and b . **2**

- (c) Consider the function $f(x) = xe^{-x}$
- | | | |
|------|---|----------|
| i) | Find the coordinates of any stationary points and determine their nature. | 2 |
| ii) | Show that there is a point of inflection at $x = 2$ | 2 |
| iii) | Hence sketch the graph of $y = f(x)$ showing all essential features. | 3 |

- (d) i) Evaluate $\int_0^{\pi} \sin^2 3x dx$ **3**
- ii) Show that $\lim_{h \rightarrow 0} \frac{1 - \cos 2h}{h^2} = 2$ **2**

- (e) MN is a tangent at R and PQRS is a cyclic quadrilateral.
PS is parallel to QR and QT is parallel to SR



Prove that:

- | | |
|--|----------|
| i) QR bisects $\angle PQT$ | 2 |
| ii) $MN \parallel PT$ | 2 |
| (f) Prove by Mathematical Induction that $2^{3n-1} + 3$ is divisible by 7 for all positive integers $n \geq 1$. | 4 |

End of Section B

Section C (Use a SEPARATE writing booklet)

Marks

Question 3 (26 marks)

- (a) An eight-person committee is to be formed from a group of 10 women and 15 men. In how many ways can the committee be chosen if the committee must contain:

i) 4 men and 4 women

1

ii) more women than men

2

iii) at least 2 women

2

- (b) Using the results

$$\frac{d(e^x \sin x)}{dx} = e^x \cos x + e^x \sin x$$

$$\text{and } \frac{d(e^x \cos x)}{dx} = e^x \cos x - e^x \sin x$$

$$\text{evaluate } \int_0^{\frac{\pi}{2}} e^x \cos x dx$$

2

- (c) A car can hold six people, three in the front and three in the back.

2

Only two of the six people can drive. In how many different seating arrangements can they complete the journey?

- (d) The portion of the curve $y = \sin x + \cos x$ between $x = 0$ and $x = \frac{\pi}{2}$ is rotated about the x -axis. Show that the volume of the solid of revolution generated is $\frac{\pi}{2}(\pi + 2)$ cubic units.

3

- (e) Using a first approximation $x_1 \neq 0$, show that the second approximation, x_2 , is such that $|x_2| > |x_1|$ when using Newton's method to obtain the zero of $\sqrt[3]{x}$

2

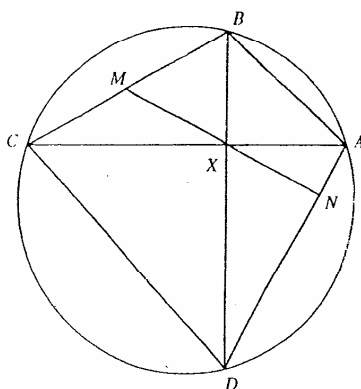
- (f) Prove by Mathematical Induction that 4

$$\cos x + \cos 3x + \cos 5x + \dots + \cos(2n-1)x = \frac{\sin 2nx}{2 \sin x},$$

for $n \geq 1$ (where n is an integer).

- (g) If $y = \frac{1}{2}(e^x - e^{-x})$, prove that $x = \log_e \left(y + \sqrt{y^2 + 1} \right)$ 3

(h)



ABCD is a cyclic quadrilateral. The diagonals AC and BD intersect at right angles at X. M is the midpoint of BC. MX produced meets AD at N. Prove that:

i) $\angle MBX = \angle MXB$ 2

ii) MN is perpendicular to AD 3

End of paper



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Mathematics Extension 1

Sample Solutions

Section	Marker
A	AMG
B	FN
C	EC

Question 1

$$(a) (i) \frac{d}{dx} \tan 2x = 2 \sec^2 2x \quad (1)$$

$$(ii) \frac{d}{dx} \left(\frac{1}{e^{x/2}} \right) = \frac{d}{dx} e^{-x/2}$$
$$= -\frac{1}{2} e^{-x/2}$$
$$= \frac{-1}{2e^{x/2}} \quad (2)$$

$$(iii) \frac{d}{dx} \ln(1-2x)^2$$
$$= \frac{1}{(1-2x)^2} \times \frac{d}{dx} (1-2x)^2$$
$$= \frac{2(1-2x) \times -2}{(1-2x)^2}$$
$$= \frac{-4}{1-2x} \quad (2)$$

$$(iv) \frac{d}{dx} \sin^2(3x)$$
$$= 2 \cos(3x) \sin(3x) \times 3$$
$$= 3 \sin(6x) \quad (2)$$

$$(b) (i) \int \cos \frac{x}{2} dx$$
$$= 2 \int \cos \frac{x}{2} dx \quad (1)$$
$$= 2 \sin \frac{x}{2} + C$$

$$(ii) \int e^{1-4x} dx = -\frac{1}{4} e^{1-4x} + C \quad (1)$$

$$(iii) \int \cot x dx = \int \frac{\cos x}{\sin x} dx$$
$$= \ln \sin u + C \quad (2)$$

$$(iv) \int 3x \cdot e^{x^2} dx = \frac{3}{2} \int 2x e^{x^2} dx$$
$$= \frac{3}{2} e^{x^2} + C \quad (2)$$

$$(c) \lim_{x \rightarrow 0} \frac{\sin 2x}{x} = 2 \lim_{x \rightarrow 0} \frac{\sin 2x}{2x}$$
$$= 2 \lim_{u \rightarrow 0} \frac{\sin u}{u}$$

(Since if $u=2x$, then $u \rightarrow 0$ as $x \rightarrow 0$)

$$= 2 \times 1$$

$$= 2 \quad (1)$$

Qn 1 contd

(d)(i) $x \cdot \ln x - 1 = 0$

When $x=1$ LHS = $1 \cdot \ln 1 - 1$
 $= 1 \times 0 - 1$
 $= -1$

When $x=2$ LHS = $2 \ln 2 - 1$
 ≈ 0.386

Since curve is continuous,
and sign changes, $\therefore$ there
is a root in $[1, 2]$ [2]

(ii) $f(x) = x \ln x - 1$

$$f'(x) = x \cdot \frac{1}{x} + 1 \cdot \ln x$$
$$= 1 + \ln x$$

$$\text{Now } z_2 = z_1 - \frac{f(z_1)}{f'(z_1)}$$

Let $z_1 = 2$

$$z_2 = 2 - \frac{f(2)}{f'(2)}$$

$$= 2 - \frac{2 \ln 2 - 1}{1 + \ln 2}$$

$$\approx 1.77$$

[2]

(2)

(e)(i) $\frac{2x+1}{x+2} = \frac{2x+4-3}{x+2}$

$$= \frac{2(x+2)-3}{x+2}$$

$$= 2 - \frac{3}{x+2}$$

[1]

(ii) $\int_0^1 \frac{2x+1}{x+2} dx = \int_0^1 \left[2 - \frac{3}{x+2} \right] dx$

$$= \left[2x - 3 \ln(x+2) \right]$$

$$= (2 - 3 \ln 3) - (0 - 3 \ln 2)$$

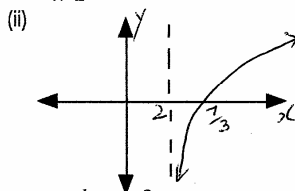
$$= 2 + 3 \ln 3 - 3 \ln 2$$

$$= 2 + 3 \ln \frac{3}{2}$$

[3]

QUESTION 2

(a) (i) $3x - 6 > 0$
 $x > 2$



(iii) $\frac{dy}{dx} = \frac{3}{3x-6}$

gradient of tangent = $\frac{1}{x-2}$

gradient of normal = $-x+2$

at $x=4$ grad of normal = -2 and $y = \log_6 6$

eqn. of normal is $y - \log_6 6 = -2(x-4)$

$2x + y - 8 - \ln 6 = 0$

(b) period = $\frac{2\pi}{b} = 8$

$b = \frac{\pi}{4}$

$3 = a \cos(b \times 1)$

$3 = a \cos \frac{\pi}{4}$

$3 = a \times \frac{1}{\sqrt{2}}$

$a = 3\sqrt{2}$

(c) (i)

$y = xe^{-x}$

$y' = e^{-x} - xe^{-x}$

$y' = e^{-x}(1-x)$

$y'' = e^{-x} - e^{-x} + xe^{-x}$

$y'' = e^{-x}(-1-1+x)$

$y'' = e^{-x}(-2+x)$

$y' = 0$ when $x=1$ and at $x=1$ $y' = \frac{1}{e}(-1)$, $y'' < 0$

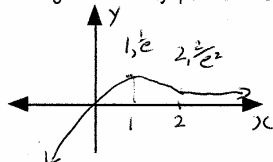
$\therefore$ max. turning point at $x=1$ (2)

(ii) $y'' = 0$ when $x=2$,

$y'' < 0$ when $x=1$ and > 0 when $x=3$ $\left(\frac{1}{e} \times 1\right)$

$\therefore$ change in concavity- point of inflexion at $x=2$ (2)

(iii)



(d) (i) $\int_0^{\pi} \sin^2 3x dx = \frac{1}{2} \int_0^{\pi} 1 - \cos 6x dx$

$= \frac{1}{2} \left[x - \frac{1}{6} \sin 6x \right]_0^{\pi}$

$= \frac{1}{2} \left[\frac{\pi}{8} - \frac{1}{6} \sin \frac{3\pi}{4} \right] - \frac{1}{2} [0 - 0]$

$= \frac{1}{2} \left[\frac{\pi}{8} - \frac{1}{6\sqrt{2}} \right]$

$= \frac{\pi}{16} - \frac{\sqrt{2}}{24}$

(3)

(ii) $\lim_{h \rightarrow 0} \frac{1 - \cos 2h}{h^2}$

$= \lim_{h \rightarrow 0} \frac{2 \sin^2 h}{h^2}$

$= 2 \times \lim_{h \rightarrow 0} \frac{\sinh}{h} \times \frac{\sinh}{h}$

$= 2 \times 1 \times 1$

$= 2$

(2)

(e) (i) Let $\angle TQR = a$

$\angle QRS = a$ (alternate $\angle$)

$\angle PSR = \pi - a$ (co-interior)

$\angle PQR = a$ (opp. $\angle$ in cyclic quad)

so PQ bisects $\angle PQT$

(2)

(ii) Join PT

from (i) $\angle PQR = \angle PTR = a$ (angles on arc PR)

$\angle RQT = \angle TRN = a$ ($\angle$ in opp. segment)

so $PT \parallel MN$ as $\angle PTR = \angle TRN$ (alt. $\angle$ s)

(2)

(f) When $n=1$, $2^{3 \times 1 - 1} + 3 = 7$

(4)

Statement true for $n=1$

Assume statement true for $n=k$

$2^{3k-1} + 3 = 7A$ where A is a positive integer.

If $n=k+1$

$2^{3(k+1)-1} + 3 = 2^{3k-1} \times 2^3 + 3$

$= 8(7A - 3) + 3$

$= 56A - 21$ which is divisible by 7

$2^{3(k+1)-1} + 3$ is divisible by 7

If the statement is true for $n=k$, it is true for $n=k+1$. So, by Mathematical Induction it is true for any integer $n \geq 1$

[Section C.]

Question (3). (a)

(i)

M (15)	W (10)

(i) $\binom{15}{4} \times \binom{10}{4} = 1365 \times 210$
 $= 286650$

(ii)

5W 3M $\binom{15}{3} \binom{10}{5} = 455 \times 252$

6W 2M $\binom{15}{2} \binom{10}{6} = 114660$
 22050

7W 1M $\binom{15}{1} \binom{10}{7} = 1800$

8W $\binom{10}{8}$ Total = 138555

(iii) 6M 2W, 5M 3W,
 4M 4W, 3M 5W, 2M 6W
 1M 7W, 8W.

(b) $\binom{15}{6} \binom{10}{2} + \binom{15}{5} \binom{10}{3} + \binom{15}{4} \binom{10}{4}$
 $+ \binom{15}{3} \binom{10}{5} + \binom{15}{2} \binom{10}{6} + \binom{15}{1} \binom{10}{7}$
 $+ \binom{10}{8}$

$= 225225 + \dots + 45$

$= 1010790$

(c)

		D

2 choices for the driver and the rest of the passengers can arrange themselves in $5!$ ways

i.e. $= 2 \times 5!$
 $= 240$

$\int (e^{x \cos x} + e^{x \sin x}) = e^{x \sin x}$

$\int (e^{x \cos x} - e^{x \sin x}) = e^{x \cos x}$

$\therefore \int_0^{\pi/2} e^{x \cos x} dx = \left[\frac{e^x}{2} (\sin x + \cos x) \right]_0^{\pi/2}$

$= \frac{e^{\pi/2}}{2} - \frac{1}{2}$

$= \frac{1}{2} (e^{\pi/2} - 1)$

(d)



$V = \pi \int_0^{\pi/2} y^2 dx = \pi \int_0^{\pi/2} (1 + \sin 2x)^2 dx$

$= \pi \left[x - \frac{\cos 2x}{2} \right]_0^{\pi/2}$

$= \left(\frac{\pi}{2} + \frac{1}{2} \right) - \left(-\frac{1}{2} \right)$

$= \frac{\pi}{2} (\pi + 2)$

Question (3).

$$(e) f(x) = x^{1/3}$$

$$f'(x) = \frac{1}{3} x^{-2/3}$$

If x_1 is the 1st approx.
then a better approx.
 x_2 is given by

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= x_1 - \frac{x_1^{1/3}}{(1/3)x_1^{-2/3}}$$

$$= x_1 - 3x_1 = -2x_1$$

$$\therefore |x_2| = |-2x_1| = 2|x_1|$$

i.e. $|x_2| > |x_1|$.

(f) Let $S(n)$ be the
statement that

$$\cos x + \cos 3x + \dots + \cos(2n-1)x$$

$$= \frac{\sin 2nx}{2 \sin x}.$$

For $n=1$,

$$LHS = \cos x$$

$$RHS = \frac{\sin 2x}{2 \sin x} = \frac{2 \sin x \cos x}{2 \sin x}$$

$$= \cos x$$

$\therefore S(1)$ is true

Assume $S(k)$ is true

Consider $n = k+1$

Now $S(k+1) =$

$$\cos x + \cos 3x + \dots + \cos(2k+1)x$$

$$= \frac{\sin 2kx}{2 \sin x} + \cos(2k+1)x$$

$$= \frac{\sin 2kx + 2 \sin x \cos(2k+1)x}{2 \sin x}$$

$$= \frac{\sin 2kx + 2 \sin x [\cos 2kx \cos x - \sin 2kx \sin x]}{2 \sin x}$$

therefore B, C, X
are concyclic pts.

$$= \frac{\sin 2kx (1 - 2 \sin^2 x) + \cos 2kx \sin 2x}{2 \sin x}$$

$$= \frac{\sin 2kx (\cos 2x) + \cos(2kx) \sin 2x}{2 \sin x}$$

$$= \frac{\sin(2kx + 2x)}{2 \sin x}$$

$$= \frac{\sin 2(k+1)x}{2 \sin x}$$

$S(k+1)$ is true if $S(k)$
is true. $\therefore$ By M.I.

$S(n)$ is true $\forall n \geq 1$
where $n \in \mathbb{Z}^+$

$$(g) 2y = e^x - \frac{1}{e^x}$$

$$2y e^x = e^{2x} - 1$$

$$\therefore (e^x)^2 - (2y)e^x - 1 = 0$$

$$\Rightarrow e^x = \frac{2y \pm \sqrt{4y^2 + 4}}{2}$$

$$e^x = y \pm \sqrt{y^2 + 1}$$

$$\because e^x > 0 \therefore e^x = y + \sqrt{y^2 + 1}$$

$$\therefore x = \ln(y + \sqrt{y^2 + 1}).$$

(h) $\triangle CXB$ is a rt. $\angle$ $\triangle$.

and $\therefore BC$ subtends
a right $\angle \Rightarrow BC$ is a
diameter.

$$\therefore MC = MB$$

$\therefore M$ is the centre

and MX is also a rad.

i.e. $\triangle MBX$ is isosceles.

$$\therefore \angle MBX = \angle MXB.$$

(ii) In $\triangle BCX$ Let $\angle B = \alpha$.

$$\therefore \angle BCA = 90 - \alpha \text{ (}\angle \text{ sum of } \triangle \text{)}$$

$$\text{and } \angle BCA = \angle BDA = 90 - \alpha$$

($\angle$ in same seg). Also

$$\angle BXM = \angle DXN = \alpha \text{ (vert. opp.)}$$

$$\therefore \angle XND = 90^\circ \Rightarrow MN \perp AD.$$